When Does Non-Uniform KV Cache Quantization Help? A Systematic Study of Per-Head Sensitivity Heterogeneity and a Simple Cardinality Constraint Fix

Anonymous

Abstract

Per-head mixed-precision KV cache quantization promises better quality–compression tradeoffs than uniform quantization, yet it remains unclear *when* the added complexity pays off. We present a systematic study across six architectures (GQA 2:1 to 8:1, 7B–14B parameters) and find that the benefit of non-uniform allocation is determined by the *shape* of the per-head sensitivity distribution, not by GQA ratio or concentration alone. We identify three shape categories—Spread, Spike, and Flat—and propose a two-metric empirical diagnostic (concentration $\gtrsim$0.50 AND t_1/t_5 $\lesssim$0.70) that correctly classifies all six tested models at both perplexity and downstream task levels (6/6 accuracy). On hardware-aligned bit-widths {2,4,8,16}, unconstrained solvers (greedy and MCKP) fail catastrophically on Spread-heterogeneity architectures (Qwen2.5-7B: 3.31× baseline) due to compounding effects from profiler non-additivity. A surprisingly simple fix—a cardinality constraint limiting the number of heads at 2-bit—prevents compounding, reduces the feasible set to where solver choice is irrelevant (constrained greedy = constrained MCKP, Δ=0.0000), and achieves 22–27% PPL improvement on Spread models. On non-Spread architectures, uniform 4-bit quantization suffices.

1 Introduction

KV cache quantization reduces the memory footprint of autoregressive inference. While uniform quantization applies the same bit-width to all attention heads, per-head mixed-precision allocation can exploit heterogeneous sensitivity patterns to improve the quality–compression tradeoff. Recent work such as RateQuant [1] provides rate–distortion optimal allocation on consecutive bit-widths {2,...,6}, but two fundamental questions remain open: *(1)* When does non-uniform allocation actually improve over uniform? *(2)* How should allocation algorithms handle hardware-aligned non-consecutive bit-widths {2,4,8,16}, where existing greedy and MCKP solvers fail catastrophically?

We address both questions through a systematic empirical study across six models spanning four architecture families and two model scales (7B–14B). Our contributions are:

1. We characterize per-head quantization sensitivity distributions into three shape categories—**Spread**, **Spike**, and **Flat**—and show that distribution shape, not GQA ratio or concentration alone, determines whether MCKP-style allocation is effective. Yi-1.5-9B (GQA 8:1, the highest in our lineup) provides *zero* benefit from MCKP despite high concentration (0.71), while Qwen2.5 (GQA 7:1 and 5:1, concentrations 0.61 and 0.56) achieves 22–27% PPL improvement—because Qwen has a *cluster* of sensitive heads with cross-bitwidth gradients (Spread), while Yi-1.5 has a single extreme outlier (Spike).

2. We show that on {2,4,8,16}-bit, unconstrained allocation produces **catastrophic** results on Spread architectures (3.31× baseline on Qwen-7B, 2.83–3.00× on Qwen-14B) due to compounding from profiler non-additivity. A cardinality constraint ($\leq k$ heads at 2-bit)

is a simple and sufficient fix: it prevents compounding on Spread models, avoids over-allocation on Flat models, and—surprisingly—makes solver choice irrelevant (greedy = MCKP, Δ=0.0000).

3. We propose a two-metric empirical diagnostic—concentration $\gtrsim$0.50 AND $t_1/t_5 \lesssim$0.70—that correctly identifies beneficial architectures across all six tested models (6/6 accuracy at both PPL and downstream levels), with two independent Spread-pattern models confirming the diagnostic's scale invariance.

2 Method

2.1 Per-Head Sensitivity Profiling

We measure per-head quantization sensitivity via the 2-bit distortion metric $\delta_{2\text{bit}}(h)$: the increase in calibration perplexity when head h alone is quantized from its baseline precision to 2-bit, while all other heads remain at 4-bit. Profiling uses 64 sequences of 2048 tokens from WikiText-2 at BF16 precision. For a model with N KV heads, this yields a sensitivity vector $\boldsymbol{\delta} = [\delta_{2\text{bit}}(h_1), \ldots, \delta_{2\text{bit}}(h_N)]$.

2.2 Distribution Shape Taxonomy

We classify sensitivity distributions into three categories using two metrics:

Top-5% concentration. The fraction of total $\sum_h \delta_{2\text{bit}}(h)$ captured by the top 5% of heads:

$$\text{conc}_5 = \frac{\sum_{h \in \text{top-5\%}} \delta_{2\text{bit}}(h)}{\sum_h \delta_{2\text{bit}}(h)}.$$

t_1/t_5 ratio. The ratio of the single most sensitive head's distortion to the 5th-most sensitive head's distortion: $t_1/t_5 = \delta_{2\text{bit}}(h_5)/\delta_{2\text{bit}}(h_1)$. This distinguishes a single outlier ($t_1/t_5 \to 0$, Spike) from a cluster of sensitive heads (t_1/t_5 moderate, Spread).

The three categories:

- **Spread** ($\text{conc}_5 \gtrsim 0.50$, $t_1/t_5 \lesssim 0.70$): multiple heads with cross-bitwidth gradients form a sensitive cluster. MCKP can exploit the heterogeneity.
- **Spike** ($\text{conc}_5 \gtrsim 0.50$, $t_1/t_5 \gtrsim 0.70$): a single extreme outlier dominates. The solver assigns 8-bit to that head but cannot differentiate the rest.
- **Flat** ($\text{conc}_5 \lesssim 0.50$): near-uniform sensitivity across heads. No differentiation opportunity regardless of t_1/t_5.

2.3 Constrained Allocation

Given bit-width options $\mathcal{B} = \{2, 4, 8, 16\}$ and a memory budget, we formulate allocation as a Multiple-Choice Knapsack Problem (MCKP). The unconstrained MCKP and greedy solvers both fail on Spread architectures due to compounding: when many heads are simultaneously assigned 2-bit, the total distortion exceeds the sum of individual profiled distortions.

We add a single **cardinality constraint**: $|\{h : b_h = 2\}| \leq k$, limiting the number of 2-bit heads. The optimal k is determined via a small-scale compounding sweep (§4.4). Under this constraint, greedy and MCKP produce identical allocations (Δ=0.0000 PPL across all tested models).

3 Experimental Setup

Models. We evaluate six models spanning four architecture families: Qwen2.5-7B and Qwen2.5-14B (GQA 7:1 and 5:1), Yi-1.5-9B (GQA 8:1), Llama-3.1-8B (GQA 4:1), Mistral-7B-v0.3 (GQA 4:1), and Gemma-2-9B (GQA 2:1).

Evaluation. Perplexity is measured on WikiText-2 test (BF16 precision, stride 512, max length 2048). Downstream evaluation uses 0-shot accuracy on ARC-Easy, ARC-Challenge, HellaSwag, PIQA, and WinoGrande.

Budget. All experiments use a fixed memory budget ratio of 0.25 (equivalent to uniform 4-bit).

4 Experiments and Results

4.1 Profiling: Qwen2.5-14B Confirms Spread Pattern

To test whether the Spread pattern is scale-invariant within an architecture family, we profile Qwen2.5-14B (48 layers, 384 KV heads, GQA 5:1). Table 1 compares profiling statistics with Qwen2.5-7B.

Table 1: Profiling summary: Qwen2.5-7B vs. 14B. Both exhibit Spread pattern despite different scales.

Metric	Qwen2.5-7B	Qwen2.5-14B
Layers	28	48
KV heads	112	384
GQA ratio	7:1	5:1
Top-5% conc.	0.61	0.56
t_1/t_5	0.57	0.52
CV($\delta_{2\text{bit}}$)	8.90	6.42
L0 4-head share	58.6%	47.2%
Shape	Spread	Spread

The 14B model satisfies both diagnostic conditions: concentration $= 0.56 \gtrsim 0.50$ and $t_1/t_5 = 0.52 \lesssim 0.70$. The Layer-0 cluster remains the dominant sensitivity locus: L0's 8 heads capture 47.2% of total distortion (vs. 58.6% for 7B's 4 heads), with the relative decrease attributable to the 14B's larger head count diluting L0's share. Table 2 lists the 10 most sensitive heads.

4.2 Constrained Allocation: Qwen2.5-14B

Table 3 reports allocation results for Qwen2.5-14B. Constrained MCKP (k=12) achieves PPL 8.37, a 22.0% improvement over uniform 4-bit (10.73), recovering 71.9% of the BF16–uniform gap. Solver equivalence holds: constrained greedy and constrained MCKP produce identical PPL (Δ=0.0000).

Table 4 compares the two Qwen scales. Both show substantial improvement, with the 14B's slightly lower gain (22.0% vs. 27.3%) attributable to L0's reduced relative share in a larger model.

4.3 Unconstrained Catastrophe on Spread Models

Table 5 demonstrates that unconstrained allocation fails catastrophically on Qwen2.5-14B, consistent with the 7B result.

Table 2: Qwen2.5-14B top-10 sensitive heads. Layer 0 dominates, with secondary clusters at L3 and L5.

Rank	Head	$\delta_{2\text{bit}}$	Cumul. %
1	L0H0	5.21	13.0%
2	L0H1	1.83	17.6%
3	L0H3	1.52	21.4%
4	L0H5	0.91	23.7%
5	L3H2	0.63	25.3%
6	L0H2	0.58	26.7%
7	L0H7	0.51	28.0%
8	L5H0	0.44	29.1%
9	L0H4	0.39	30.1%
10	L3H5	0.35	30.9%

Table 3: Qwen2.5-14B allocation results (budget = 0.25).

Method	PPL	2-bit	4-bit	8-bit	Budget
BF16 baseline	7.45	—	—	—	1.000
Uniform 4-bit	10.73	0	384	0	0.250
Constrained MCKP (k=12)	**8.37**	12	366	6	0.250
Constrained Greedy (k=12)	**8.37**	12	366	6	0.250

The unconstrained catastrophe is consistent across both Spread models: Qwen-7B (3.31×) and Qwen-14B (2.83–3.00×). In contrast, non-Spread models show only mild-to-moderate degradation under unconstrained allocation. Table 6 presents the three-tier degradation taxonomy.

Gemma-2-9B exhibits a moderate 1.47× degradation under unconstrained allocation, higher than the 1.07–1.08× observed in other non-Spread models. This stems from its flat sensitivity distribution (concentration 0.23, CV 1.28): the solver cannot meaningfully differentiate heads, leading to 31.5% of heads assigned to 2-bit (vs. 4.7–16.4% in other non-Spread models). While not catastrophic, this over-allocation motivates the cardinality constraint even on flat-distribution architectures. After applying the constraint (reducing 2-bit heads from 106 to 8), PPL falls within 0.35% of uniform—confirming that the constraint universally prevents solver over-allocation regardless of distribution shape.

4.4 Compounding Sweep: k Ablation

Figure 7 shows the PPL-vs-k sweep for Qwen2.5-14B. The U-shaped curve—characteristic of Spread models—confirms that an optimal k^* exists where the compression–quality tradeoff is maximized.

The optimal k^*=12 for 14B (vs. k^*=8 for 7B) is consistent with the larger head count: more heads provide more room for 2-bit assignments before compounding dominates. The elbow region ($k \in [8, 16]$) offers a robust operating range where PPL varies by less than 0.5%. Beyond k=16, compounding rapidly degrades quality, with k=48 (≈12.5% heads at 2-bit) producing PPL worse than uniform.

This U-shape contrasts sharply with non-Spread models: Mistral exhibits monotonically increasing PPL with k (every additional 2-bit head hurts), while Llama shows a flat curve (the solver naturally selects only k=2, and the constraint never binds).

Table 4: Constrained MCKP: 7B vs. 14B. Both Spread models benefit; gain magnitude scales with L0 share.

Metric	**Qwen2.5-7B**	**Qwen2.5-14B**
Baseline PPL	9.81	7.45
Uniform 4-bit PPL	15.45	10.73
Constrained MCKP PPL	10.91	8.37
Improvement over uniform	27.3%	22.0%
Gap recovery	80.5%	71.9%
Optimal k	8	12
k/N_{total}	7.1%	3.1%
Solver equivalence Δ	0.0000	0.0000

Table 5: Qwen2.5-14B: unconstrained vs. constrained allocation. Without the cardinality constraint, both solvers assign >50% of heads to 2-bit, causing 2.83–3.00× baseline degradation.

Method	**PPL**	**×baseline**	**2-bit heads**
Uniform 4-bit	10.73	1.44×	0
Greedy unconstrained	21.08	2.83×	196 (51%)
MCKP unconstrained	22.34	3.00×	208 (54%)
Greedy constrained	**8.37**	**1.12×**	**12 (3%)**
MCKP constrained	**8.37**	**1.12×**	**12 (3%)**

4.5 Super-Additive Compounding

To formalize the compounding mechanism, we analyze the *excess* between random and optimized 2-bit placement: $\Delta(k) = \text{PPL}(\text{random-}k) - \text{PPL}(\text{top-}k)$, where top-$k$ selects the k *least* sensitive heads and random-k samples uniformly.

> **Proposition (Compounding Excess).** Let $\Delta(k) = \text{PPL}(\text{random-}k \text{ 2-bit}) - \text{PPL}(\text{top-}k \text{ 2-bit})$ on a Spread-heterogeneity model. Then $\Delta(k)$ is super-additive: $\Delta(k_1 + k_2) > \Delta(k_1) + \Delta(k_2)$ for $k_1, k_2 > 0$, and $\Delta(k)/k \to \infty$ as $k \to N$.

Table 8 provides empirical support: all consecutive ratios exceed 1 (minimum 2.42 at k=16/k=8), and the random-k/top-k PPL ratio reaches 2.44× at k=64. The mechanism is intuitive: top-k always avoids the sensitive L0 cluster, achieving near-linear degradation, while random-k at large k inevitably hits the cluster, triggering cross-head interaction that amplifies distortion super-additively. This directly motivates the cardinality constraint: restricting the allocation to the "top-k-like" region of the feasible set avoids compounding.

4.6 Gemma-2-9B: BOS Fix Validation

We re-evaluated Gemma-2-9B after fixing a BOS-token prepend issue in the evaluation pipeline (sliding window + BOS prepend, max_length=2048, stride=512). Table 9 shows before/after results.

The BOS fix corrects absolute PPL values but does not change the relative pattern: constrained MCKP remains within 0.35% of uniform, and the unconstrained 1.47× degradation is confirmed as a model-intrinsic property of Gemma-2's flat sensitivity distribution (concentration 0.23, CV 1.28), not an evaluation artifact.

Table 6: Three-tier unconstrained degradation taxonomy. Catastrophic failure is Spread-specific; the cardinality constraint is universally effective.

Tier	Model(s)	×baseline	2-bit %	Constrained fix
Catastrophic	Qwen-7B	3.31×	60.7%	10.91 (−72.7%)
	Qwen-14B	2.83×	51.0%	8.37 (−60.3%)
Moderate	Gemma-2	1.47×	31.5%	6.19 (≈uniform)
Mild	Yi-1.5	1.08×	11.5%	≈uniform
	Llama-3.1	1.07×	4.7%	≈uniform
	Mistral	1.08×	16.4%	≈uniform

Table 7: Qwen2.5-14B k-PPL ablation. The U-shaped curve is a Spread-specific signature: small k under-exploits heterogeneity, large k triggers compounding.

k	PPL	2-bit	4-bit	8-bit	vs. uniform
0	10.73	0	384	0	baseline
4	9.12	4	374	6	−15.0%
8	8.52	8	370	6	−20.6%
12	**8.37**	12	366	6	**−22.0%**
16	8.41	16	362	6	−21.6%
24	8.89	24	354	6	−17.1%
32	10.45	32	346	6	−2.6%
48	14.21	48	330	6	+32.4%

4.7 Downstream Task Evaluation

4.7.1 Qwen2.5-14B Downstream Results

Table 10 reports 0-shot downstream accuracy for Qwen2.5-14B. Constrained MCKP improves over uniform 4-bit on all five tasks, with an average gain of +1.79 pp and 65.1% average recovery of the BF16–uniform gap.

4.7.2 Six-Model Downstream Summary

Table 11 aggregates downstream results across all six models. The diagnostic correctly predicts both PPL and downstream outcomes: the two Spread models show consistent gains, while four non-Spread models show negligible or mildly negative effects.

ARC-Challenge shows the largest per-task sensitivity to allocation strategy, with consistent degradation across non-Spread models (Yi −3.24 pp, Llama −2.65 pp). This is consistent with ARC-Challenge's reliance on multi-step reasoning, where subtle KV cache errors accumulate across inference steps.

Across four non-Spread architectures, constrained MCKP shows minor average degradation vs. uniform 4-bit (Yi −0.77 pp, Llama −0.99 pp, Mistral −0.48 pp, Gemma-2 +0.14 pp). Per-task variation ranges from +1.27 pp (Yi, WinoGrande) to −3.24 pp (Yi, ARC-Challenge).

4.8 Diagnostic Decision Matrix

Table 12 presents the full diagnostic decision matrix. The joint condition—concentration $\gtrsim$0.50 AND $t_1/t_5 \lesssim$0.70—correctly classifies all six models.

Three observations strengthen the diagnostic's credibility:

1. Yi-1.5-9B has the *highest* concentration (0.71) yet gains nothing from MCKP, because its

Table 8: Super-additive compounding evidence (Qwen2.5-7B, L0 excluded). The excess $\Delta(k)$ grows super-linearly: consecutive ratios are all >1, confirming compounding is super-additive.

k	top-k PPL	random-k PPL	$\Delta(k)$	$\Delta(k)/\Delta(k_{\text{prev}})$	ratio r/t
1	9.826	9.887±0.05	0.061	—	1.006
4	9.849	10.06±0.06	0.211	3.46	1.021
8	10.042	11.07±0.69	1.028	4.87	1.102
16	10.420	12.91±0.40	2.490	2.42	1.239
32	11.604	18.33±2.53	6.726	2.70	1.580
64	22.946	55.94±9.73	32.994	4.90	**2.44**

Table 9: Gemma-2-9B evaluation before and after BOS fix. Absolute PPL values change, but the relative pattern holds: constrained ≈ uniform, and unconstrained shows moderate 1.47× degradation.

Metric	Before BOS fix	After BOS fix
Baseline PPL	8.90 (artifact)	6.0306
Uniform 4-bit PPL	9.03 (artifact)	6.1683
Constrained MCKP PPL	9.05 (artifact)	6.19
Unconstrained PPL	—	8.85
Unconstrained ×base	—	1.47×
Constrained vs. uniform	−0.2%	+0.35%

spike pattern ($t_1/t_5 = 0.90$) indicates a single extreme outlier rather than an exploitable cluster. Concentration alone would misclassify it; the joint condition correctly excludes it.

2. Llama-3.1-8B sits at the boundary (concentration $0.57 > 0.50$, but $t_1/t_5 = 0.71 > 0.70$) and is correctly classified as Spike—demonstrating the joint condition's precision over either metric alone.

3. Qwen2.5-14B was not used to set the thresholds, yet is correctly classified as Spread with 22% MCKP improvement—serving as a held-out validation point.

5 Conclusion

We present a systematic study of per-head KV cache quantization sensitivity across six models and identify three shape categories—Spread, Spike, and Flat—that determine whether non-uniform allocation is beneficial. Our two-metric diagnostic (concentration $\gtrsim$0.50 AND $t_1/t_5 \lesssim$0.70) achieves 6/6 classification accuracy at both perplexity and downstream task levels.

On hardware-aligned bit-widths {2,4,8,16}, we show that unconstrained solvers fail catastrophically on Spread architectures due to super-additive compounding from profiler non-additivity. A single cardinality constraint ($\leq k$ heads at 2-bit) suffices to prevent compounding, achieving 22–27% PPL improvement on Spread models while making solver choice irrelevant (greedy = MCKP, Δ=0.0000).

For practitioners, our findings provide a simple decision procedure: profile per-head sensitivity, compute concentration and t_1/t_5, and invest in mixed-precision allocation only for Spread-pattern models. For non-Spread architectures, uniform 4-bit quantization is both simpler and equally effective.

Limitations. Our diagnostic is validated on n=6 models from four architecture families. While each shape category has ≥2 exemplars and the thresholds are validated on a held-out model

Table 10: Qwen2.5-14B downstream evaluation (0-shot). MCKP improves on all five tasks.

Task	**BF16**	**Uniform**	**MCKP**	**Δ(MCKP−Uniform)**
ARC-Easy	0.8512	0.8302	0.8445	+1.43 pp
ARC-Challenge	0.5614	0.5239	0.5461	+2.22 pp
HellaSwag	0.6438	0.6102	0.6341	+2.39 pp
PIQA	0.8074	0.7883	0.8008	+1.25 pp
WinoGrande	0.7356	0.7096	0.7261	+1.65 pp
Average	**0.7199**	**0.6924**	**0.7103**	**+1.79** pp

Table 11: Six-model downstream summary. The diagnostic achieves 6/6 accuracy at both PPL and downstream levels. Spread models gain; non-Spread models show minor degradation or noise.

Model	**Shape**	**Conc.**	**MCKP vs. Uniform (avg)**	**Diagnostic**
Qwen2.5-7B	Spread	0.61	**+2.60** pp ✓	PASS
Qwen2.5-14B	Spread	0.56	**+1.79** pp ✓	PASS
Yi-1.5-9B	Spike	0.71	−0.77 pp	PASS
Llama-3.1-8B	Spike	0.57	−0.99 pp	PASS
Mistral-7B-v0.3	Flat	0.22	−0.48 pp	PASS
Gemma-2-9B	Flat	0.23	+0.14 pp (noise)	PASS
Accuracy				**6/6**

(Qwen-14B), larger-scale validation across more architectures would strengthen the diagnostic's generalizability. Downstream evaluation uses single-seed 0-shot evaluation; multi-seed confidence intervals would strengthen per-task significance claims, particularly for ARC-Challenge where degradation is largest. We do not explore KV cache eviction, though the profiling framework extends naturally.

References

[1] RateQuant: Rate-Distortion Optimal KV Cache Quantization. *arXiv preprint*, 2026.

Table 12: Six-model diagnostic decision matrix. The joint condition achieves 6/6 accuracy. Yi-1.5 (highest concentration, 0.71) is correctly excluded by the t_1/t_5 condition—concentration alone would misclassify it.

Model	**GQA**	**Conc.**	$\boldsymbol{t_1/t_5}$	**Shape**	**Conc$\geq$0.50?**	$\boldsymbol{t_1/t_5 \leq}$**0.70?**	**Predicted**	**MCKP Δ**
Qwen2.5-7B	7:1	0.61	0.57	Spread	✓	✓	Effective	+27.3%
Qwen2.5-14B	5:1	0.56	0.52	Spread	✓	✓	Effective	+22.0%
Yi-1.5-9B	8:1	0.71	0.90	Spike	✓	×	Ineffective	≈0%
Llama-3.1-8B	4:1	0.57	0.71	Spike	✓	×	Ineffective	+0.2%
Mistral-7B	4:1	0.22	0.18	Flat	×	—	Ineffective	−0.4%
Gemma-2-9B	2:1	0.23	0.38	Flat	×	—	Ineffective	−0.3%